

THARUN PRANAV SAKTHIVEL

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EDUCATION

University of British Columbia, Vancouver, BC

Expected April 2027

• BSc, Combined Major in Science (Statistics, Physics and Environmental Sciences)

RELEVANT EXPERIENCE

Berribot, AI Engineer, San Francisco, CA | [Case Study](#)

Jan. 2026 – Apr. 2026

- Architected Berribot's JD-to-candidate ranking pipeline with BM25, dense retrieval, RRF, Qwen3, and LambdaMART to accelerate recruiter screening, cutting shortlisting time by 42% across the workflow.
- Designed recruiter-labelled ranking evals with nDCG@k, MRR, Recall@k, and W&B to validate search quality before rollout, benchmarking 100,000 JD-resume pairings to guide ranking and retrieval changes.
- Spearheaded BerriTutor, a voice-first AI tutor using LiveKit, WebRTC, Supermemory, and a multi-agent harness to personalize live learning while supporting 4K+ concurrent tutoring sessions.
- Implemented a Langfuse evaluation stack for tracing, prompt versioning, and 800+ LLM-as-a-judge tests to catch model regressions before release, cutting manual review from 4+ hours to minutes.
- Deployed 4+ production services with Docker, GitHub Actions, GKE, and Cloud Run to keep AI workloads reliable at peak demand, maintaining 99.8% uptime while handling 5K+ interviews in 24 hours.
- Engineered a resume intelligence pipeline using extraction, normalization, and skill taxonomy mapping to standardize candidate profiles for search and ranking, increasing downstream match precision by 38%.
- Built MediaPipe pipeline to track candidate movement and flag cheating, reducing manual interview review.

UBC Agrobot, AI Engineer (Engineering Design Team), Vancouver, BC

Oct. 2023 – Jan. 2026

- Fine-tuned YOLO on 8,000+ field images, improving weed recall from 80% to 89% maintaining precision.
- Trained a U-Net to guide robot steering, cutting steering offset from 12 cm to 5 cm in field tests.
- Analyzed weed detection errors in lighting and occlusions, retrained hard cases and cut false negatives by 28%.

Hyr, AI Engineer, Singapore | [Case Study](#)

Jan. 2024 – Jul. 2024

- Engineered multi-stage LLM interview agents using OpenAI function calling, cutting screening time by 70%.
- Built candidate profiling with resume parser using spaCy, pgvector, and composite scoring, serving 10K+ users.
- Shipped an async Celery pipeline using Pydantic validation for JD generation, reducing role setup time by 80%.

RK>, Software Engineering Intern, Dayton, OH | [Case Study](#)

Jun. 2023 – Aug. 2023

- Built NLTK semantic search with custom NER to surface relevant PDF content using vector and entity scores.
- Engineered Python RFID localization to track 400+ tagged products across 8 office zones for inventory use.
- Modeled log-distance RSSI across 40 landmark tags to improve RFID localization with 2K+ labeled samples.

PERSONAL PROJECTS

ThirdSlate, AI-Powered Learning Platform

[Case Study](#) [Source Code](#)

- Built grounded RAG with LangGraph and Gemini to reduce hallucinations, achieving 89% RAGAS faithfulness.
- Implemented multimodal ingestion with PyMuPDF and PaddleOCR, cutting document processing time by 60%.
- Architected 3 FastAPI services to keep ingestion and quiz generation from blocking RAG, serving 1K+ users.

Tracebox, Voice Agent Testing Platform

[Case Study](#) [Source Code](#)

- Designed an 8-state LangGraph agent to automate browser voice tests across reasoning, speech, and audio.
- Built a WebRTC proxy to test voice agents by capturing speech and feeding 16 kHz TTS as virtual mic input.
- Engineered RMS silence detection to stop after speech, preventing voice tests from stalling beyond 120s.

TECHNICAL SKILLS

- **Backend:** Python, TypeScript, FastAPI, Node.js, Next.js, PostgreSQL, Redis, Celery, REST, WebSockets, OAuth
- **AI & ML:** LLMs, RAG, AI Agents, PyTorch, Hugging Face, LangGraph, Pydantic, MCP, Langfuse, pgvector, RAGAS
- **Cloud & DevOps:** Docker, Kubernetes, GCP, AWS, GKE, Cloud Run, GitHub Actions, CI/CD, OpenTelemetry, Sentry